

Chapter 4: Control Flow

The Decision-Making Heart of Trading Algorithms

On October 22, 2015, a relatively quiet autumn day on Wall Street, High-Frequency Trading (HFT) firm Virtu Financial accomplished something that seemed almost magical to traditional traders. In just a single trading day, across thousands of securities and dozens of global markets, they completed the day with positive trading results—making money rather than losing it—for the 1,237th consecutive trading day.

This streak was so remarkable that it caught the attention of the entire financial world. No human trader in history had ever achieved such consistency. What was Virtu's secret?